



CNN-IBN's Rocking Affair with PostgreSQL

Developers and companies that build or deploy software solutions for websites that have to be available 24/7 can rely on PostgreSQL as a cost-effective alternative to proprietary enterprise database solutions. In this article, EnterpriseDB PostgreSQL contest winner Manish Singh reveals how he used PostgreSQL to deal with downtimes for CNN-IBN's website.

Managing news websites is no trivial task, and it can cost a company millions if the website goes down for even a minute. A live news website should always be well equipped to handle a sudden increase in traffic or load, server problems related to power failures and even hard-disk crashes. During such events, the main challenge is to continue streaming data to website viewers without interruption so that users remain unaware about the backend problems. If the website shows up such errors to the viewer, it results in turning away hundreds (if not thousands) of visitors while reducing the effectiveness of purchased ad campaigns and associated revenue. *Ibnlive.in.com*, one of Network 18's most reputed news websites, faced such problems, and gradually resolved them by implementing PostgreSQL servers along with a Memcached server and a load balancer.

Challenges

Ibnlive.com witnessed a massive growth in the number of website viewers, but it did not foresee the technical problems that this growth brought. The high load of traffic was tough to handle, and as a result, the website started experiencing

frequent downtimes. As a media website, it had to prepare for high-traffic days and felt the need for a replication feature. When the database server went down due to power failure, the website did not display news stories that viewers requested, and instead, displayed error messages to them. "One of the worst problems we faced was when a hard-disk server blew out, and the website went down for over two hours—at a time when we were expecting high traffic," reveals Manish Singh, senior Linux administrator, CNN-IBN. For a live news website that gets 8-9 million views a day, a downtime of two hours is unacceptable. He continues, "We took a few hours to gather the data that was stored on the server, but there were many corrupted files. After deleting those, we managed to retrieve only 70 per cent of the files stored. For two hours and ten minutes, there were no stories on the website, and we lost many clients, revenue and couldn't fulfil our purchased ad campaigns, as the advertisements were not displayed during the downtime."

A stand-by solution

Memcached 1.2.5 was installed as a 'middleman' server for the website. This object-caching system saves repeated database queries by caching

'answer' data when it is obtained from the database server(s). Hence, when a visitor to the site tries to view a story, the website sends a request to the *Memcached* server. If the required data is available in cache, the *Memcached* server supplies it, saving a database query. In case the data is not cached by the *Memcached* server, it acts as an intermediary and requests the required data from other servers. Responses to the *Memcached* server are passed back to the website. "This way, even if a database server fails, the website continues to communicate smoothly with the viewer, without any interruptions," comments Singh.

Singh's team also set up a stand-by server (log archiving, in other words), by which the server gets updates from other main servers every ten minutes, and stores updated 16 MB files using *Wal* (Write ahead log) archives. The log describes every change made to the database's data files. This log exists primarily for crash-safety purposes, and if the system crashes, the database can be restored by replaying the log entries made since the last checkpoint. Stories that formed the website's content were stored in different directories. When a main server